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CMOS-Memristor based LIF & STDP functions for Spiking Neural Networks Application

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***Abstract*—This paper presents a hybrid analog spiking neural network (SNN) architecture that combines a threshold switching memristor (TSM)-based leaky integrate-and-fire (LIF) neuron with a CMOS-implemented spike-timing-dependent plasticity (STDP) circuit. The proposed circuit generates biologically realistic spike behavior and drives synaptic learning by modulating the resistance of non-volatile memristors. TaOx and AIST-based memristive devices are employed as synapses to investigate material-dependent learning characteristics. Simulation results confirm that the TSM-LIF neuron exhibits tunable firing patterns, and the STDP circuit successfully adjusts synaptic weights according to spike timing. A comparative analysis reveals that AIST-based memristors demonstrate greater sensitivity to control voltage than TaOx counterparts. The architecture offers a modular, scalable, and power-efficient solution for neuromorphic hardware learning.**

***Index Terms*—Memristor, SNN, TSM, LIF, STDP**

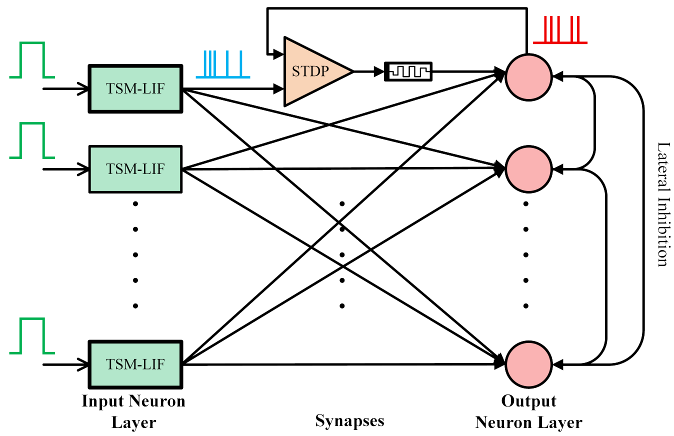

Fig. 1: Simplified SNN architecture

I. Introduction

Spiking Neural Networks (SNNs) emulate biological information processing by using discrete spikes to transmit and compute data, offering a pathway to brain-inspired hardware with massive parallelism and ultra-low power consumption [1], [2]. Unlike traditional artificial neural networks that rely on continuous-valued activations, SNNs are driven by spike timing, making them well-suited for real-time and event-driven computation in neuromorphic systems [3], [4].

At the heart of most SNNs lies the leaky integrate-and-fire (LIF) neuron model, which captures essential features of biological neurons such as membrane integration, threshold-based firing, and refractory behavior [1], [3], [5]. Its low hardware complexity and strong biological grounding make LIF a popular choice for analog and mixed-signal neuromorphic circuits [4]. However, conventional CMOS implementations of LIF neurons often suffer from scalability and power challenges due to their high transistor count [2], [6].

To mitigate these limitations, researchers have explored integrating memristive devices into LIF neuron circuits. Memristors offer nanoscale footprints, non-volatility, and dynamic switching behavior, which enables compact and energy-efficient implementations of both neuron and synapse functions [1], [7]. Volatile threshold switching memristors (TSMs), in particular, have been shown to naturally replicate the integrate-and-fire behavior of biological neurons without the need for complex auxiliary circuits [3], [5], [8]. These properties allow TSM-based neurons to reduce area and power consumption while maintaining functional fidelity.

In parallel, spike-timing-dependent plasticity (STDP) has emerged as a key unsupervised learning rule in neuromorphic systems. By adjusting synaptic weights based on the relative timing of pre- and post-synaptic spikes, STDP enables hardware systems to learn spatiotemporal patterns in an online, local manner [4], [6]. Memristive devices have been widely adopted for implementing STDP-based synapses, due to their analog resistance modulation and temporal dynamics [7], [9]. However, direct integration of learning behavior into memristive synapses often reduces tunability and complicates design due to the variability and limited controllability of emerging devices [9].

To address this, CMOS-based circuits have been proposed for generating STDP learning signals externally, providing precise control over timing windows and update behavior [4], [6]. These circuits can compute the learning voltage based on spike timing and apply it to external non-volatile memristive synapses, enabling hybrid learning architectures with both flexibility and efficiency. Notably, recent works have also explored scalable analog implementations of LIF and STDP for low-power systems [6], while others have demonstrated the feasibility of modular synapse designs with improved signal integrity and learning resolution [9].

In this work, we propose a novel hybrid neuromorphic architecture shown in Fig. 1 that leverages a TSM-based LIF

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neuron to generate precise pre- and post-synaptic spikes, which are then fed into a CMOS-based STDP circuit. This STDP circuit computes a control voltage that modulates the memristance of non-volatile memristive synapses. By separating the learning logic from the memory device, the system enhances programmability, reliability, and modularity. Specifically, the work originality relies on the following three core contributions:

1) The integration of a volatile threshold switching memristor (TSM)-based leaky integrate-and-fire (LIF) neuron with a CMOS STDP circuit to provide biologically realistic and programable spike-based learning.
2) Architectural decoupling of memory and learn functionality to provide exact and programable synaptic updates without locking in complexity to the memristive device itself.
3) A comparative analysis of TaOx and AIST memristive devices based on the same paradigm of learning, which can provide insights on their quality for neuromorphic applications.

This mixed-hardware architecture with a direct module approach represents a scalable and energy-efficient path for real-time, analog SNN hardware realizations.

This paper is organized as follows: analog circuit implementation of TSM-LIF & STDP circuit is shown in Section II, Section III presents the simulation results of the circuits followed by the conclusion in Section IV.

II. Analog Circuit Implementation

A. Design of TSM-LIF Circuit

The TSM-based LIF neuron replicates core functions of biological neurons by accumulating input over time, simulating membrane potential decay through charge leakage, and emitting a spike upon reaching a defined threshold. While simplified, it effectively models the temporal firing patterns observed in real neurons.

Fig. 2 illustrates the structural analogy between a biological neuron and its electronic counterpart implemented using a TSM. The TSM used here has been modeled in Verilog-A using parameters derived from practical device behavior as described by *Liu et al.* in [10]. The following equations describes the behavior of TSM:

$$\begin{aligned} V_{pn} &= I_{pn} * \left[R_{on} * \frac{x}{D} + R_{off} * \left(1 - \frac{x}{D}\right)\right] \\ x &= x_{last} + \Delta x, \quad 0 \le x \le D \\ \Delta x &= \begin{cases} K_{on} * (V_{pn} - V_{th}), & V_{pn} > V_{th} \\ K_{off} * (V_{hold} - V_{pn}), & V_{pn} < V_{hold} \\ 0, & V_{hold} \le V_{pn} \le V_{th} \end{cases} \end{aligned} \tag{1}$$

here V_{pn} represents the voltage across the TSM devices, while I_{pn} denotes the corresponding current. The devices operate with distinct resistance states: R_{on} for the conductive (on) state and R_{off} for the non-conductive (off) state. The parameter D defines the range of variability between these two extremes. x is the internal state variable that corresponds to memristance. At every interval during the simulation, the change in memristance, noted as Δx, updates the previous resistance state x_{last} to form the current state x, which is simply the sum of x_{last} and Δx. The dynamics of switching are controlled by unit-less coefficients K_{on} and K_{off}, which are calibrated to reflect realistic switching speeds of the TSMs. When V_{pn} surpasses the threshold voltage V_{th}, Δx progressively increases until x reaches the upper bound D. Conversely, if V_{pn} drops below the holding voltage V_{hold}, Δx decreases steadily until the resistance state returns to zero.

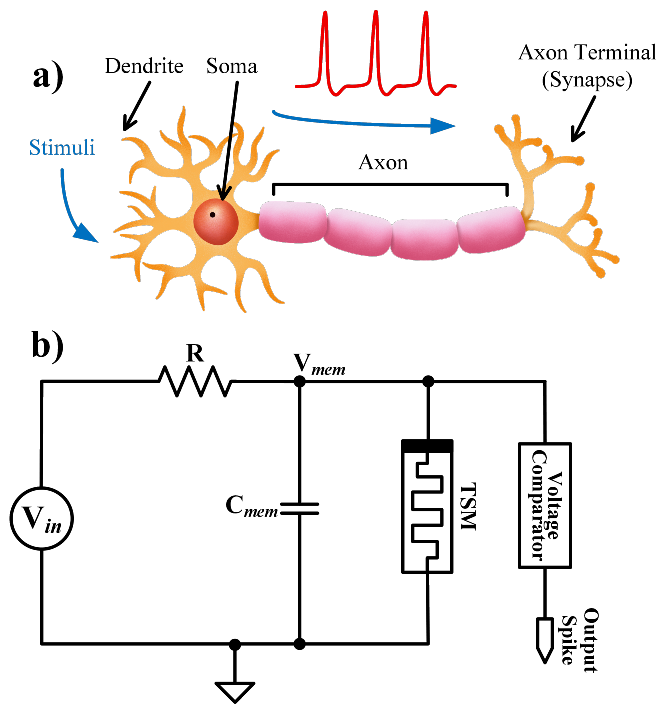

Fig. 2: Structural comparison between (a) a biological neuron and (b) its electronic equivalent using a TSM-based LIF circuit.

The LIF circuit consists of an RC charging circuit connected in parallel with the TSM. Initially, the TSM remains in a high-resistance state until the membrane voltage V_{mem} surpasses the defined threshold. Upon reaching this threshold, the TSM switches to a low-resistance state, quickly discharging the accumulated charge from C_{mem}, resulting in a current spike that mimics neuronal firing.

To convert this current spike into a voltage-readable output, a comparator circuit (Fig. 2b) is used. This voltage output is essential for further digital interfacing and downstream STDP processing. The temporal characteristics of the neuron—such as firing rate and integration delay—can be tuned by adjusting the values of R and C_{mem}, providing a controllable emulation of biological time constants.

B. Design of STDP Circuit

STDP is a form of Hebbian learning where the timing difference between pre- and postsynaptic spikes determines

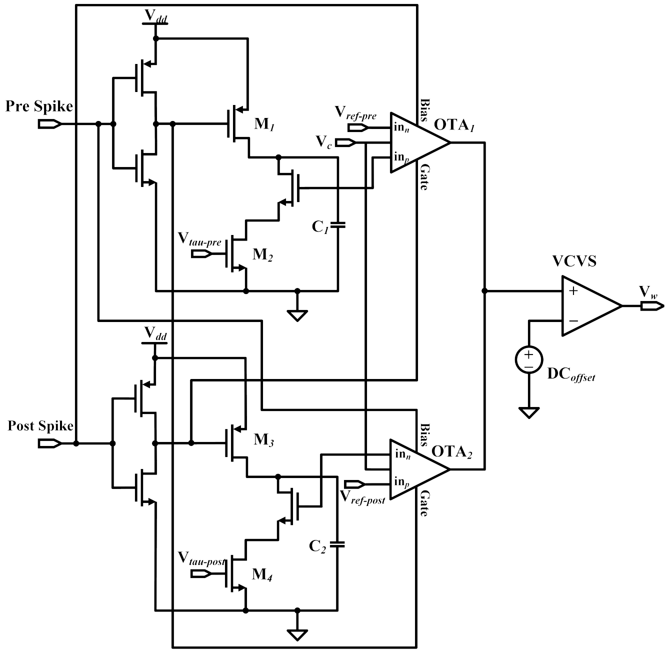

Fig. 3: STDP circuit

the direction and magnitude of synaptic weight change. If the presynaptic neuron fires before the postsynaptic one ($\Delta t < 0$), the synapse is potentiated (LTP), whereas if the postsynaptic neuron fires first ($\Delta t > 0$), the synapse undergoes depression (LTD) [11].

STDP mechanism can be explained using these two equations:

$$\Delta w = \begin{cases} A^{+} \exp\left(-\frac{\Delta t}{\tau_{+}}\right), & \Delta t < 0 \\ -A^{-} \exp\left(-\frac{\Delta t}{\tau_{-}}\right), & \Delta t > 0 \end{cases} \tag{2}$$

where Δw represents change in synaptic weight, $\Delta t = t_{post} - t_{pre}$ time difference between the post-synaptic and pre-synaptic spike, if $\Delta t < 0$ that is pre-synaptic spike precedes post-synaptic spike Δw increases and if $\Delta t > 0$ that is pre-synaptic spike follows post-synaptic spike Δw decreases, A^{+} maximum magnitude of positive weight change, $-A^{-}$ maximum magnitude of negative weight change, τ_{+} & τ_{-} are time constants that control how fast the weight change decays with timing difference.

Considering the upper half of the symmetric STDP circut, when the pre-synaptic spike transitions to a high voltage and passes through the inverter, M_1 transistor becomes conductive, allowing capacitor C_1 to charge rapidly. Once the pre-synaptic signal ceases and returns to a low voltage, C_1 begins a gradual discharge via M_2 transistor, thereby initiating the operational transconductance amplifier (OTA), depicted in Fig. 3. This circuit operates under a 1.2V supply (V_{dd}). The interaction between bias and gate signals governs the OTA's activity. When a post-synaptic spike occurs, the bias is set high and the gate low, prompting OTA_1 to raise the voltage Vw. The voltage chnage persists solely for the duration of the post-synaptic spike and ceases immediately afterward. Additionally, the circuit design allows the time constants τ_{+} and τ_{-} in Equation 2 to be tuned via $V_{tau-pre}$ and $V_{tau-post}$, as these voltages dictate the discharge behavior of capacitors C_1 and C_2.

The design uses a Voltage Controlled Voltage Source (VCVS) at the output of OTA's to generate a programmable V_w that serves two purposes:

1) Amplifying the spike-timing difference into a measurable analog voltage, and
2) Ensuring V_w can swing in both positive and negative directions, which is necessary to program memristive devices with symmetric threshold voltages.

A gain of 1.8 and a DC offset of 586mV were used to center V_w around the required programming window for both AIST and TaOx synapses. This configuration allows consistent activation of the resistive switching behavior in the memristors during training cycles.

III. Simulation Results

A. LIF Circuit Simulation

The SNN circuit was designed and simulated using 45nm CMOS process technology in the Cadence Virtuoso simulation environment. Fig. 4 shows the neuron's spiking behavior in response to various input step voltages. A delay of 2.35ms was set via the RC time constant.

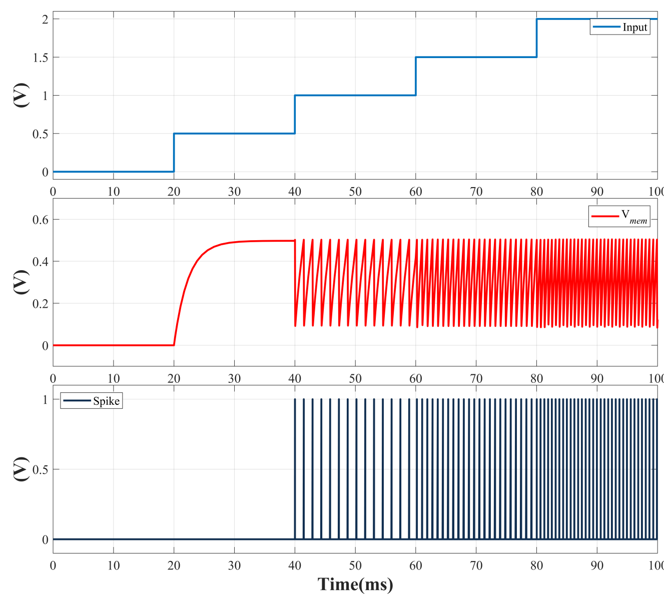

Fig. 4: Response of TSM-LIF neuron circuit to different input step voltages

As seen in Fig. 4, the TSM does not activate below 0.5V, and V_{mem} does not completely reset to 0V after each spike. This partial discharge is consistent with the volatile nature of TSMs, which rely on hold voltage thresholds to reset their resistance state. This automatic return to the high-resistance state following a spike eliminates the need for explicit reset circuitry [12].

Additionally, the spiking frequency was observed to scale with input amplitude, confirming that input intensity modulates neuron firing rate—similar to biological rate coding.

B. SNN Circuit Simulation

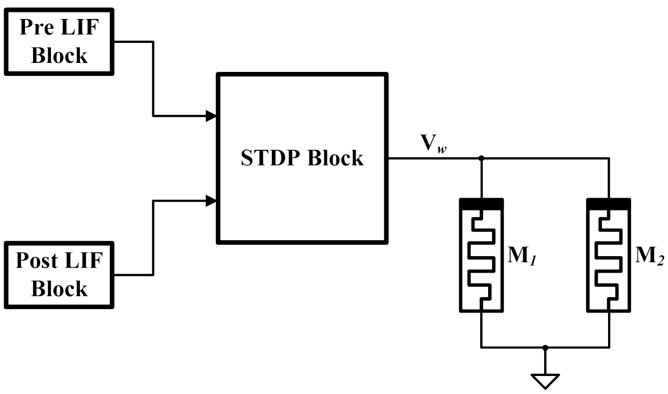

Fig. 5: Simulation setup of the hybrid SNN architecture. Two TSM-based LIF neuron circuits act as pre- and post-synaptic nodes, with their output spikes routed into a CMOS-based STDP learning circuit. The control voltage V_w generated by STDP modulates the resistance of two memristive synapses

Fig. 5 presents the setup of a complete SNN circuit core used for testing the learning dynamics. It consists of two TSM-LIF blocks acting as pre- and postsynaptic neurons. Pre- and post-synaptic spike trains were generated using Pulse Width Linear (PWL) signals and applied to the STDP block.

After evaluating the time difference Δt between spikes, the STDP circuit generates the control voltage Vw, which is then applied to two memristors acting as synapses. M_1 is based on TaOx characteristics, and M_2 on AIST [7].

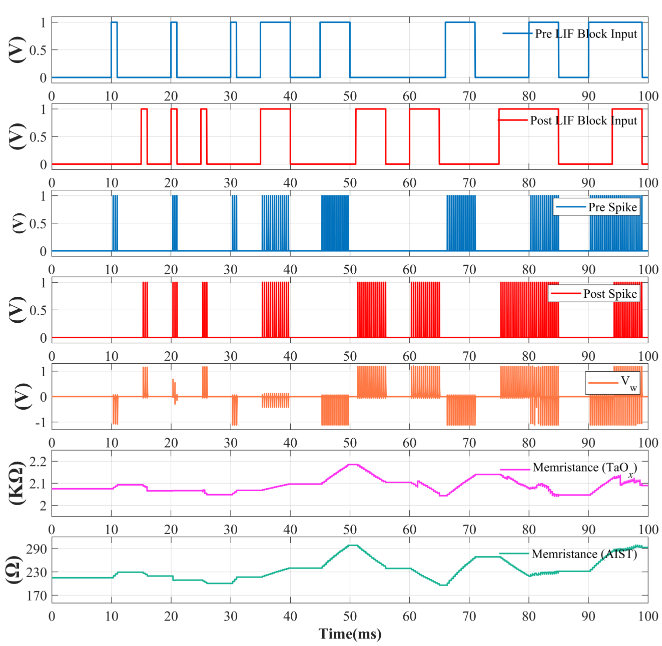

Fig. 6: Response of the SNN circuit

The rationale for using two different memristive materials is to compare how material properties affect learning behavior. Fig. 6 shows that AIST-based M_2 demonstrates a sharper and more sensitive response to V_w compared to TaOx-based M_1, suggesting that AIST may be more suitable for fine-grained learning applications.

Both synapses exhibited stable and directionally consistent resistance changes with respect to the applied control voltage, validating the effectiveness of the modular hybrid architecture.

Due to the lack of native support for memristive devices in standard process design kits (PDKs), it is currently not feasible to carry out full benchmarking and physical layout implementation of SNNs incorporating memristors. However, recent works, such as *Maheshwari et al.* [13], have demonstrated methodologies to manually integrate memristor models into commercial EDA environments like Cadence. This includes custom layout creation, device recognition for DRC/LVS checks, and evaluation of area, power, and energy consumption metrics. Therefore, while the current limitations prevent immediate implementation, the development of custom design flows and layout practices, as detailed in [13], provides a foundational direction for future benchmarking and layout exploration in memristor-based SNN architectures.

IV. Conclusion

This work presents a compact and energy-efficient neuromorphic architecture that combines TSM-based LIF neurons with a CMOS-based STDP learning circuit and non-volatile memristive synapses. The decoupling of spiking, learning, and memory components enhances the system's modularity, design flexibility, and control over learning dynamics. Through simulation, the LIF circuit demonstrated tunable firing behavior, while the STDP block produced accurate, spike-timing-dependent learning responses.

The use of both TaOx and AIST-based memristive synapses enabled a comparative evaluation of material-dependent behavior, revealing that AIST devices exhibit greater sensitivity to the applied control voltage. This suggests that material selection plays a critical role in achieving precise and efficient learning responses, an important consideration in hardware-based SNN design.

Overall, the proposed architecture demonstrates the feasibility of developing scalable, low-power SNN hardware with tunable learning capabilities and flexible synaptic design. It provides a solid foundation for future work in integrating emerging memory devices into practical neuromorphic systems.

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